

6(a)(i)	P: receptor ; Q: adeny(yl) cyclase ; R: G-protein ; S: cyclic AMP / cAMP ;	4
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Question	Answer		Marks										
6(a)(ii)	<table><tr><th>process</th><th>effect on rate of process when blood glucose concentration decreases</th></tr><tr><td>glycogenolysis</td><td>increase</td></tr><tr><td>glycogenesis</td><td>decrease</td></tr><tr><td>glycolysis</td><td>decrease</td></tr><tr><td>fatty acid synthesis</td><td>decrease</td></tr></table>	process	effect on rate of process when blood glucose concentration decreases	glycogenolysis	increase	glycogenesis	decrease	glycolysis	decrease	fatty acid synthesis	decrease	;;;	3
	process	effect on rate of process when blood glucose concentration decreases											
	glycogenolysis	increase											
	glycogenesis	decrease											
	glycolysis	decrease											
	fatty acid synthesis	decrease											
4 correct = 3 marks 3 correct = 2 marks 1/2 correct = 1 mark													
6(b)	any four from: 1 strip contains glucose oxidase and peroxidase ; 2 strip dipped into urine ; 3 glucose (and oxygen) reacts with glucose oxidase to produce hydrogen peroxide ; 4 hydrogen peroxide reacts with peroxidase to produce a colour change ; 5 colour is matched with a colour chart to give, reading / estimate, of glucose concentration ; any three from: 6 quick(er) ; 7 (more) sensitive / accurate / precise ; 8 gives immediate reading ; 9 results can be stored electronically / can be connected to smart phone / AW ; 10 numerical / digital / quantitative / objective ; 11 reusable ;		7										